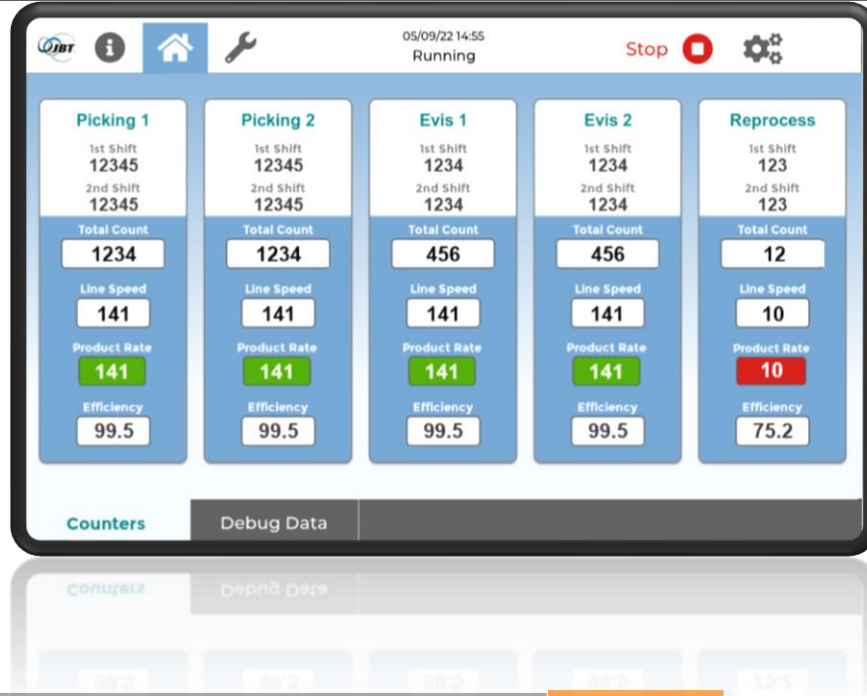




MARS
FOOD PROCESSING SOLUTIONS



JBT 1ST PROCESSING COUNTERS



Customer:

Sales Order #:

Serial #:

1	General Information
2	Operations and Instructions
3	Sanitation
4	Maintenance and Troubleshooting
5	Assembly Drawings
6	Controls and Electrical Schematics
7	Spare Parts
8	Technical Part Information

Safety Instructions & Technical Manual

Prepared by:

Engineer Approval:

Approved by:

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GENERAL INFORMATION

SAFETY GUIDELINES



DANGER!

WARNING!

All operations, maintenance and sanitation personnel **MUST** read and understand this manual and become familiar with the equipment before operating, maintaining or sanitizing.

Operators **MUST REMOVE** all **RINGS, WATCHES, NECKTIES, LOOSE GARMENTS,**

and any other items that might become entangled in the moving parts of the machine.

NO CHAIN MESH GLOVES should be worn while operating this machine.

Operators **MUST KEEP HANDS AWAY** from the **CUTTING AREA** and any **ROTATING** or **MOVING PARTS** at all times.

This machine is equipped with operating safety interlocks, which must be fully functional at all times. No attempt should ever be made to bypass safeties or run the machine with any guard removed.

Do not attempt to service machine unless power has been disconnected and locked out/ tagged out, and the machine has come to a complete stop.

NOTE: Exceptions to OSHA 29 CFR 1910.147 – Control of Hazardous Energy (Lockout/Tagout Standard) are as follows below.

Minor tool changes, adjustments and other minor service activities that take place during normal production are not included in the Lockout/ Tag out Standard provided they are:

- Routine, repetitive and integral to the use of the equipment
- Performed using alternative measures that provide effective protection for the employee



Failure to heed these warnings and comply with the instructions for safe operation of this machine can and will cause serious bodily injury and or death.

GENERAL INFORMATION

SYSTEM DESCRIPTION

System Description

The **JBT First Processing Counter System** supports a maximum of eight counters with a single JBT controller. **JBT Counters** measure the efficiency of an operation by counting product as it comes down a line and then comparing the actual count against a projected count. Counters also provide a line speed and product rate per minute so that personnel can monitor and try to keep the speeds at the desired rate. JBT Counters are utilized in many areas of a poultry processing facility. Examples include counting birds on kill lines, evisceration lines, salvage lines and debone lines as well as counting thighs or leg quarters that are being processed by an automatic deboning machine. JBT Counters also support external lights that can be used to alert personnel when the desired product rate drops to an unacceptable level or when a line is stopped unexpectedly.

The words “photoeye” and “sensor” are used interchangeably in this manual.

Options

1 – 8 Counters (Product & Shaft Sensors)

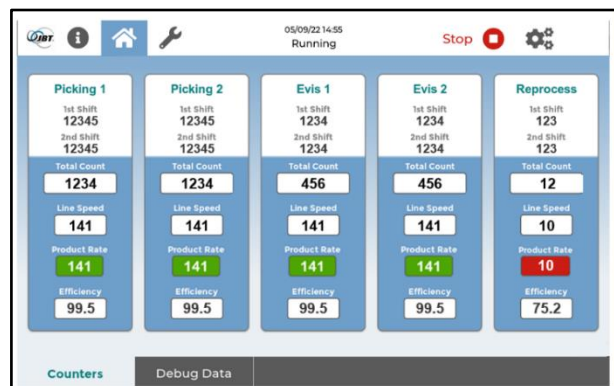
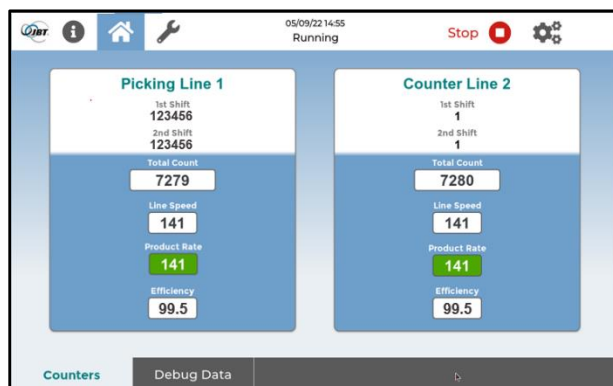
Optional Redundant Counters

Optional external lights for product rate, no product lights or line down

Optional Lot Change Tracking

JBT Efficiency Management System

Most counter installations also include JBT EMS (Efficiency Management System) software that runs on a Windows computer in a nearby office. The JBT EMS software continuously uploads data during the day and stores it in a Microsoft SQL Server database. The program makes this data available to users via reports, charts, status screens, printing, automatic report emailing, alarms and many other useful features. There is also a client version of the software that allows users to run the program on their own computer in their office.



GENERAL INFORMATION

EQUIPMENT TRAINING RECORD

All Sanitation and Maintenance personnel must be trained in the following areas.

Prime Equipment Group recommends you keep a record of personnel training.

Prime Equipment Group is willing to assist with training, following a request to do so.

Please feel free to call Prime Equipment Group for additional information.

- Installation and product application
- Disassembly and sanitation
- Instruction and parts manual
- Daily periodic maintenance
- Wearing parts
- Assembly and setup for operation
- Safety features and precautions
- Troubleshooting
- Recommended spare parts inventory
- Routine preventative maintenance schedule

GENERAL INFORMATION

LIMITED WARRANTY

Products purchased from Manufacturer are warranted only by the Manufacturer's warranty that the Product provided herein is of their standard quality and operable and that the Product will be free from defects in workmanship and material. The warranty period, for Product sold by Manufacturer and not expressly made subject to a different warranty, is one hundred twenty (120) days from date of receipt by Customer. There is no insurance against or warranty of any type for production loss due to late or damaged Product shipments. The Manufacturer makes no express or implied statutory warranties other than as expressly set forth herein.

Manufacturer shall for a period of one hundred twenty (120) days from the date any Product sold hereunder is received by Customer, repair or replace at its option, free of charge and F.O.B. point of manufacture, any nonconforming or defective parts or Product that upon inspection by Manufacturer, are deemed to be nonconforming or defective. Manufacturer shall, however, only be required to repair or replace such parts or Products if Customer (i) owned the nonconforming or defective parts or Product continuously from the original date of delivery, (ii) stored and maintained such parts and Product in accordance with Manufacturer's recommendations and standard industry practices and (iii) notified Manufacturer in writing of any non-conformance or defective parts or Product within fifteen (15) days following the date the nonconformity or defect became reasonably apparent. Further the obligations of Manufacturer hereunder do not apply to the repair or replacement of Products within the aforementioned warranty period for (a) misuse, negligence or accident, (b) normal maintenance services, (c) normal replacement of service items in connection with such service, (d) normal deterioration due to wear and tear, (e) use of non- manufacturer (OEM) replacement parts. The remedy provided in this Warranty shall be the sole and exclusive remedy of Customer in the event of defective or nonconforming parts or Product, and shall be void if the Product has been altered in any manner or has been moved from its original location without the written approval of the Manufacturer. The repair or replacement of any parts or Products under the foregoing provision does not extend the warranty beyond the warranty period described above. All electric motors, controls and other electrical services and their component parts, hydraulic motors, controls and other hydraulic services and their component parts, pneumatic controls, pneumatic services and other pneumatic components, bearings, gearboxes etc. are not included within this Warranty; the warranty with respect to these parts being limited to the manufacturer of these parts, a copy of which may be obtained from Manufacturer upon request. Customer will be responsible for the shipping/packaging costs associated with the safe and expedient return of warranted parts. The Warranty can only be claimed if genuine Manufacturer spare parts are used for repairs and maintenance of the Product.

MANUFACTURER'S UNDERTAKINGS AS CONTAINED IN THIS WARRANTY AS SPECIFIED ABOVE SHALL BE IN LIEU OF ANY OTHER GUARANTY OR WARRANTY, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY IMPLIED WARRANTY OF MERCHANTABILITY OR IMPLIED WARRANTY OF FITNESS FOR A PARTICULAR PURPOSE. IN ADDITION, IT IS EXPRESSLY AGREED BETWEEN THE PARTIES THAT THE MANUFACTURER HAS NOT MADE ANY REPRESENTATIONS OR WARRANTY AND THERE IS NO AGREEMENT THAT THE PRODUCT PURCHASED HEREUNDER WILL MEET ANY PARTICULAR

STANDARD UNLESS CUSTOMER HAS SUPPLIED MANUFACTURER WITH WRITTEN SPECIFICATIONS OF THE PRODUCT AND THE MANUFACTURER HAS AGREED IN WRITING THAT THE PRODUCT WILL MEET SPECIFIC STANDARDS SPECIFIED BY THE CUSTOMER. CUSTOMER UNDERSTANDS AND ACKNOWLEDGES THAT THERE ARE NO WARRANTIES WHICH EXTEND BEYOND THE TERMS HEREOF UNLESS OTHERWISE NOTED IN PURCHASE AGREEMENT.

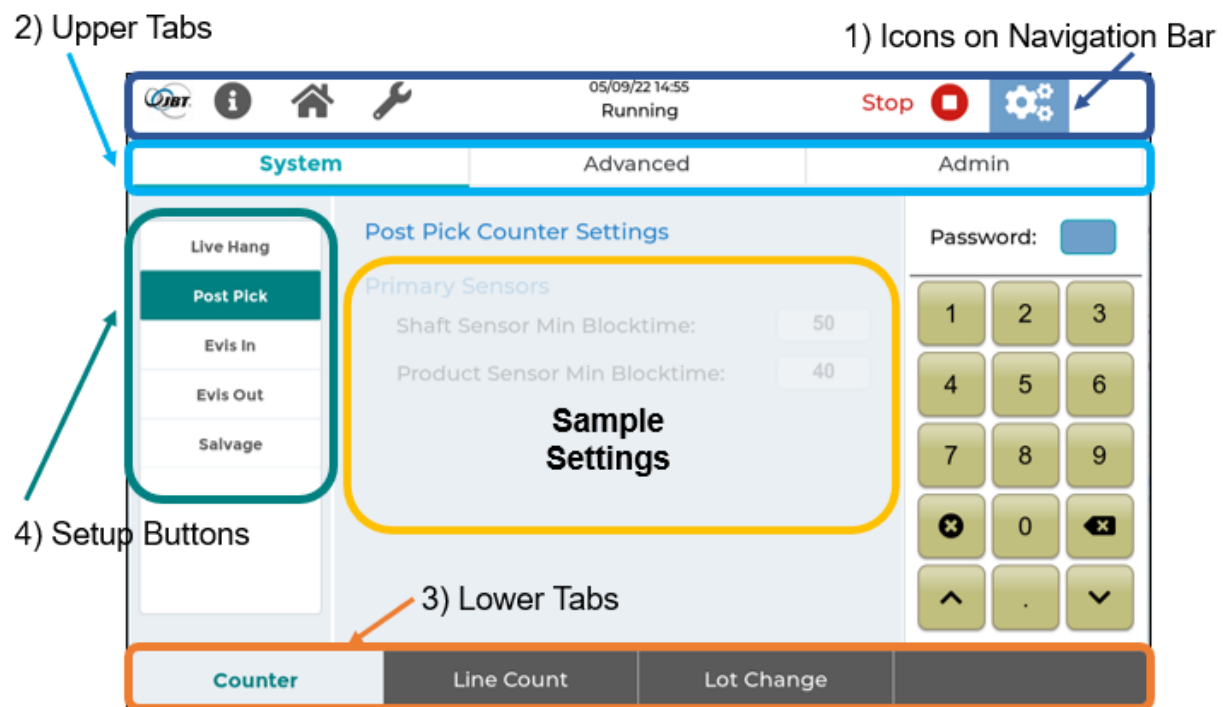
NO CLAIM OF ANY KIND, INCLUDING, BUT NOT LIMITED TO CLAIMS OF NEGLIGENCE AND BREACH OF CONTRACT, WHETHER AS TO PRODUCT DELIVERED OR SERVICES RENDERED OR FOR NON- DELIVERY OF PRODUCT OR NON-PERFORMANCE OF SERVICES SHALL BE GREATER THAN THE PURCHASE PRICE OF THE PRODUCT OR SERVICES IN RESPECT OF WHICH THE CLAIM IS MADE. WITHOUT LIMITING THE FOREGOING, MANUFACTURER'S LIABILITY HEREUNDER SHALL, UNDER NO CIRCUMSTANCES, EXCEED THE PURCHASE PRICE SET FORTH IN THIS QUOTATION. MANUFACTURER SHALL, IN NO EVENT, BE LIABLE FOR ANY INCIDENTAL, CONSEQUENTIAL, INDIRECT OR SPECIAL DAMAGES, LOSS OF BUSINESS PROFIT, PERSONAL INJURY, BUSINESS INTERRUPTION AND LOSS OF BUSINESS, EVEN IF MANUFACTURER HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES AND ANY CLAIM BY CUSTOMER TO SUCH DAMAGES, WHETHER BASED ON TORT, CONTRACT OR OTHERWISE IS HEREBY EXPRESSLY WAIVED AND EXCLUDED.

JBT System Navigation

Navigation Overview

Navigation occurs in the following steps. Some screens may not use each of these steps (for example, the screen may not need setup buttons on the left), however the steps are always in this order.

Sample navigation steps to a setup screen:

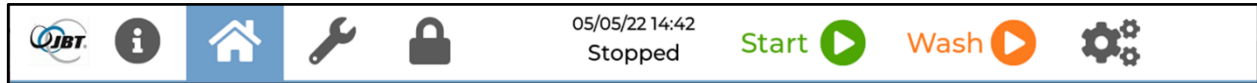


Settings (Cogs) – **System** (Upper Tab) – **Counter** (Lower Tab) – **Post Pick** (Setup Button)

OPERATIONS AND INSTRUCTIONS

NAVIGATION BAR

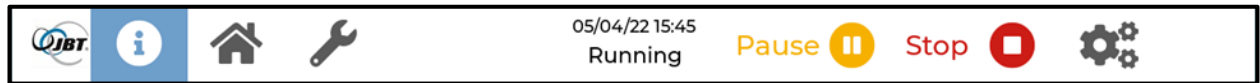
Navigation Bar



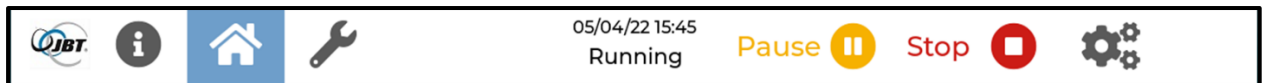
The Navigation Bar is the 1st layer of navigation. The selected icon is shown in blue.

Icon	Name	Description
	Info (About)	Info about the company, contact info, program name and version.
	Home	The home screen shows the system in action. Startup and time-outs default to this screen.
	Operator Tools	The Tools section contains tools commonly used by operators. (Operator Settings, Hardware Test, Scale Calibration, etc)
	Lock Screen	Lock the system (optional icon)
05/04/22 15:45 Running	System State	The center of the navigation bar shows the current time and system state
Start	Start	Start the System. There is an option on Controller Setup to Autostart after Power-Up.
Stop	Stop	Stop the System. Passcord required to stop the system.
	Settings	The Settings section contains settings that are generally not used in everyday operations. System Settings, Advanced Settings, Security Settings, Admin Settings, etc.

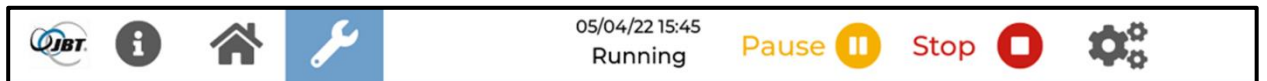
Navigation Site Map



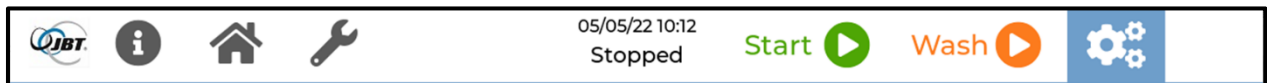
- **Info/About**
 - Company
 - Contact Info
 - Software Version



- **Home**
 - **System Console**
 - Counter
 - Debug
 - Show Block Times, Compare Sensors, Trolley Debug Data



- **Tools**
 - Reset Counters
 - Hardware Test
 - Line Count Data



- **Settings**
 - **System**
 - Counter
 - Line Count
 - Lot Change (Optional)
 - **Advanced**
 - Controller
 - I/O List
 - **Admin**
 - Security
 - General
 - JBT Adv Admin (JBT Technician)
 - Adv Counter Settings (JBT Technician)

Navigating Setup Screens

- 1) Enter password.
- 2) If accepted, the screen will replace the password with a Save Button.
- 3) Navigate to the desired field by touching the field or using the up/down symbols on the keypad.
- 4) The **Save Button** will display a red border when changes have been made.

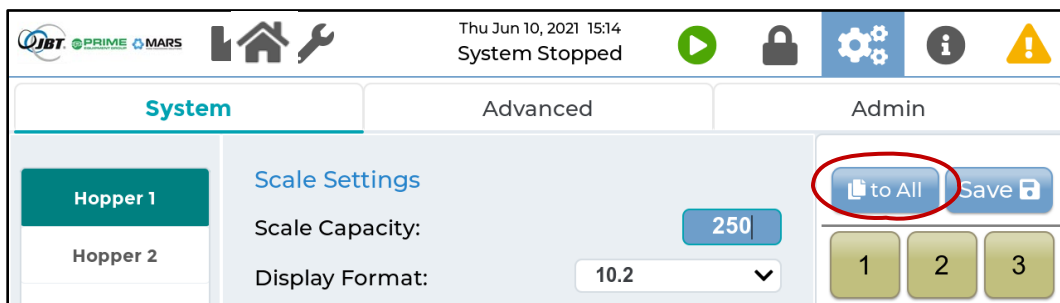
The red border is an alert that the Save Button must be pressed before leaving the screen. Changing to a different screen cancels any changes and logs the user out of the current screen.

Note: Side tabs (Counter 1, Counter 2, etc.) load a different element on the same screen, thus changing a side tab does **NOT** cancel changes and log the user out of the current screen.

For example: If user loads Hopper Settings and updates a field, then switches to look at Hardware Test without saving, the changes on Hopper Settings will be canceled.

Compared to Gen 1: Gen 1 setup screens contained a 'Back' button, which if pressed returned to the previous screen without saving changes. In Gen 2, a 'Back' (cancel) command is performed when changing to another screen without saving. Gen 1 'Save' button saves the changes and exits the screen. Gen 2 'Save' button saves the changes and does not exit the screen.

- 5) **Copy to All button** copies the values of the 1st entity to other entities.



Checkbox touch areas include the label. Touching any part of the checkbox/label line is usually sufficient to toggle the checkbox.

Gen 1 vs Gen 2

Software interface changes can be challenging – even good changes take some time to adjust. The Gen 2 interface retains as much of the original foundation as possible. Here are some notes that might make it easier to transition from the Gen 1 interface to the Gen 2 interface.

Home = System Console

All the screens previously on the System Console are still on the System Console. The only difference is that Gen 1 touch screen toggles have been replaced with tabs across the bottom of the System Console screen. For example, in Gen 1 if the Totals Report was accessed by touching the system console, Gen 2 will use a bottom tab called 'Totals'. **Bottom Tabs replace the touch toggles between System Console screens.**

Tools = Operator Tools

Screens/Tools that Operators use most frequently are located under the wrench icon. (Operator Settings, Calibrate Scales, Hardware Test, etc.). **Normal day-to-day activity is conveniently located on the System Console and the Tools section.**

Settings (Cogs) = System Setup = System Configuration

This collection focuses on system configuration, whereas Tools focuses on system operation. Most of the settings screens require passcodes higher than Operator level. The System settings are split into 'System' and 'Advanced', matching the Gen 1 options of 'System Setup' and 'Advanced Setup'. **The options under System and Advanced are the same between Gen 1 and Gen 2.** Admin Settings are covered under Gen 2 New Features.

Info = Contact Info

New In Gen 2

- The Navigation Bar at the top allows navigation directly to a selection, and returns to that selection. For example, if a user has navigated to Tools -> Hardware Test, then touches Home to return to System Console, then touches Tools again, the user returns to Hardware Test.
- The structure of the setup screens allows the user to go directly to the needed setup screen without using 'Next' to navigate thru unused screens.
- The Start / Stop buttons allow the system to be Started / Stopped from any location. Gen 1 required a return to System Console to stop the system.
- The Lock icon allows a plant to lock the screen on demand.
- Hardware Test updates periodically to show output activity in real time.
- Scale Calibration is all done on one screen.
- Scale Calibration screen can be used to test scale weights. (Password and recalibration not required to view scale weights at the bottom of the screen.)

New Admin Settings (Under Settings/Cogs Icon)

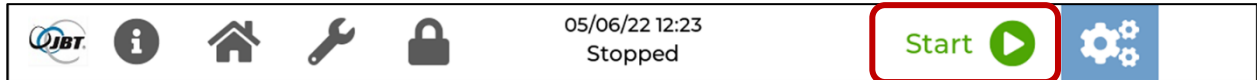
- **Security - Tiered Security**
 - The plant can manage their own passwords (for up to 4 unique passwords).
 - Logins are based on hierarchy of Operator -> System Settings -> Advanced. System Setting passcodes will work on Operator Tasks, Advanced passwords will work on Operator and System Settings.
- **I/O List** The I/O list of your input and output numbers.
- **General Admin** Set Time, Exit Program and other misc general tasks.

OPERATIONS AND INSTRUCTIONS

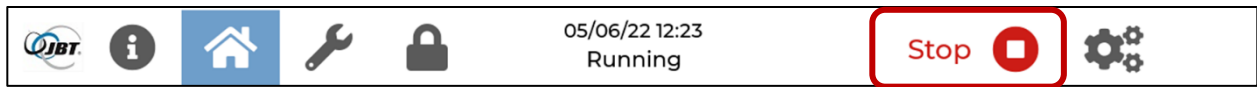
SYSTEM CONSOLE - SYSTEM CONTROLS

System Console – System Controls

System controls are located on the navigation bar.



Start the system by touching the **Start** button on the **Navigation Bar**. The system is also started automatically if a button isn't touched within 20 seconds after powering up the **JBT Controller**. An option to enable/disable autostart on power-up is on Settings (Cogs) – Advanced (Top Tab) – Controller (Bottom Tab)



Touch the Stop icon to stop the system. A Level 1 Operator or above password is required to stop the system.

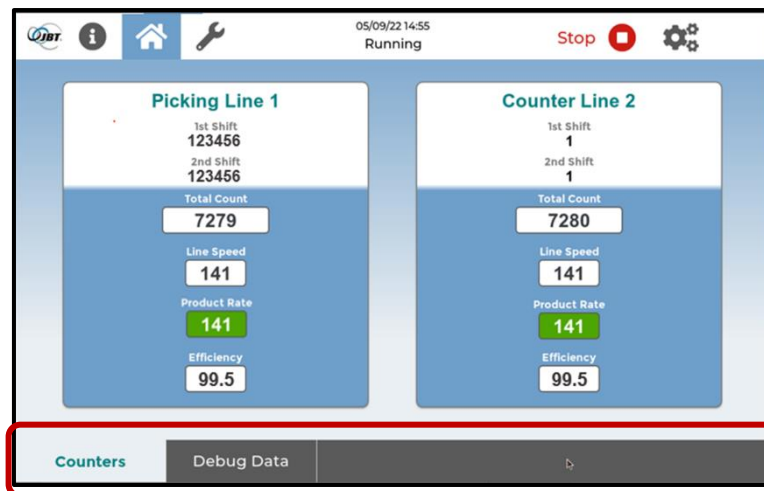
Adjusting Settings

Touch the **Tools (Wrench)** icon to access the **Operator Tools** such as Bird Reject Button activation, Hardware Test, or Line Count Setup.

Touch the **Settings (Cogs)** icon for system configuration settings..

It is not necessary to stop the system to adjust the settings.

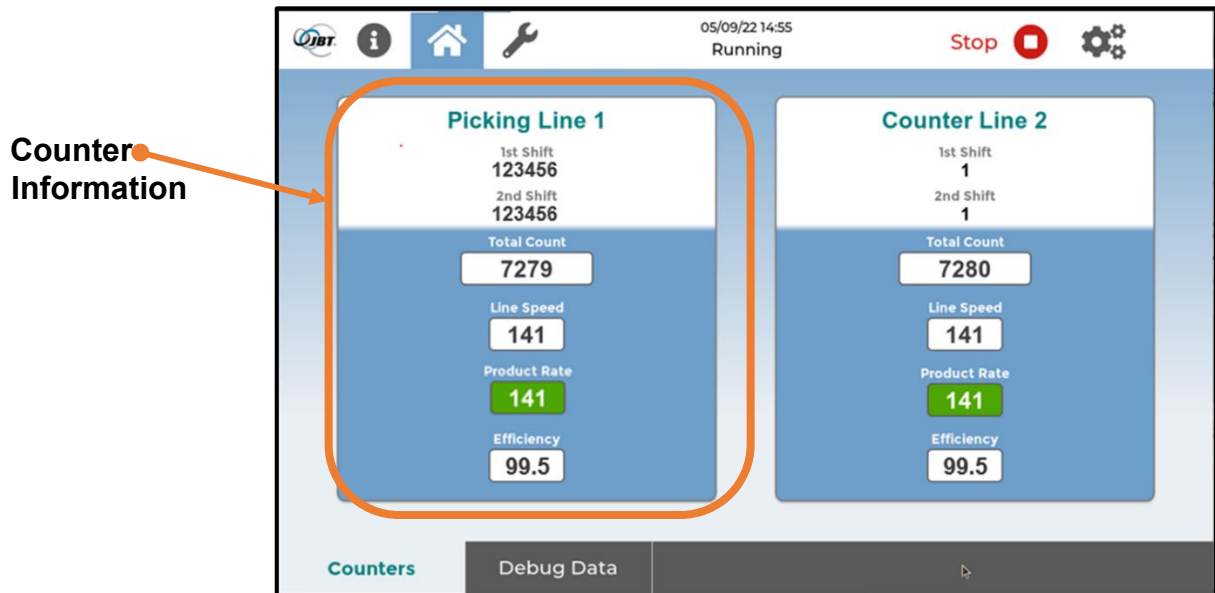
System Console Screens



Screens associated with the system console/home screen are accessed by the tabs across the bottom of the home screen

System Console – Counter Screen

The **1st Processing System Console** displays the shift count, total count, line speed, product rate and product efficiency for each counter. The System Console can show up to eight counters.



Total Count: The total number of products (birds) counted for the current shift.

Line Speed: The number of shackles counted for the previous minute.

Product Rate: The number of shackles with product (a bird) on them that were counted for the previous minute. The product rate will be displayed with a green background if the rate is within the acceptable range or with a red background if out of range

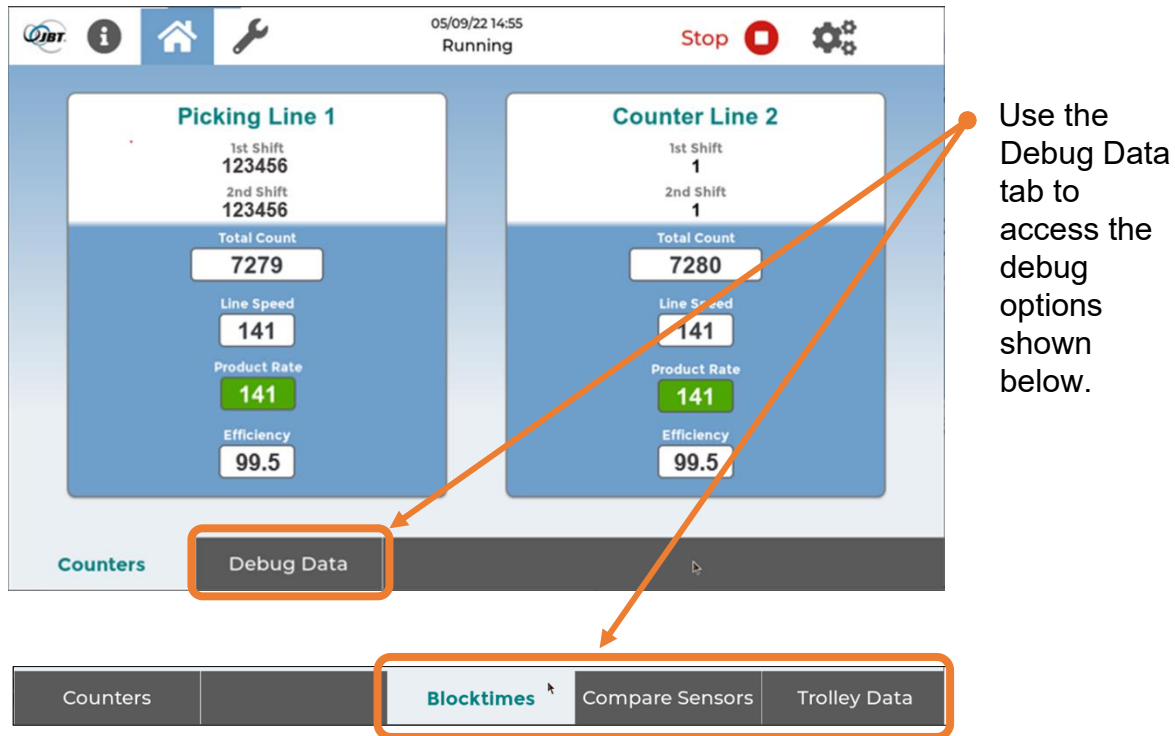
Product Efficiency: The percentage of shackles (or cones) that had birds on them for the current shift.

OPERATIONS AND INSTRUCTIONS

DEBUG DATA

Debug Data

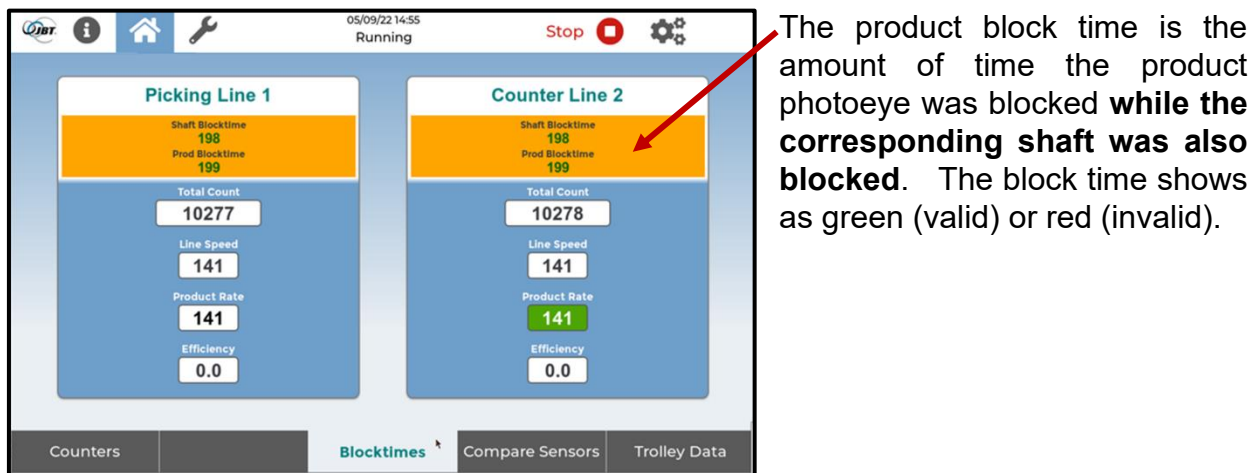
The **Debug Data** tab provides a login to access debug data screens.



Touching the Counters tab or the Home icon cancels the Debug Screens.

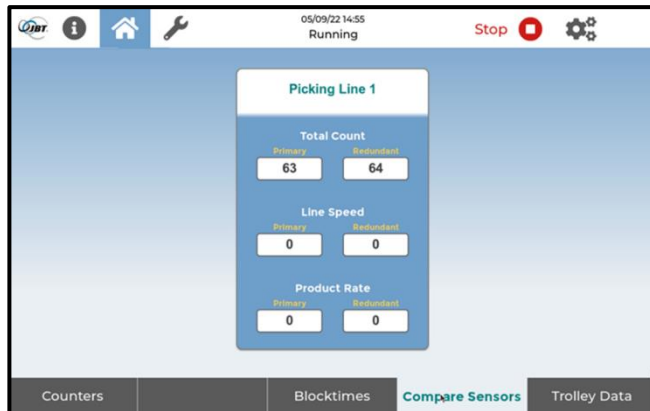
Show Block Times

The block time durations are shown in the yellow section below.



Compare Sensors

Compare the counts of the primary and redundant sensors. This feature only applies to counters with redundant sensors.

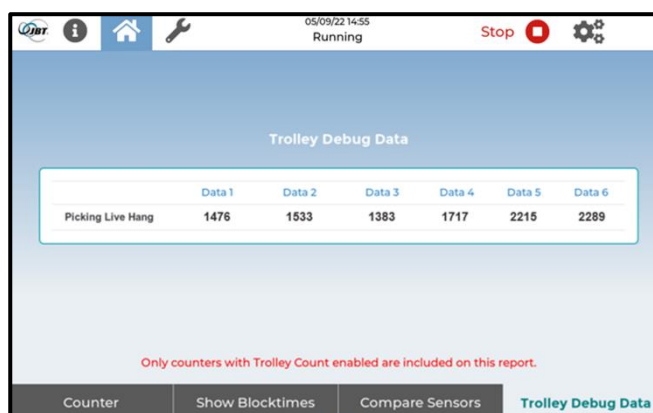


For each counter with redundant sensors, the **Primary Count** is displayed with the **Compare Count** below. The Primary Count is the current product (bird) count for the primary sensors and the Compare Count is the current product count for the redundant sensors.

Typically, the Redundant Count should be within 0.4% of the Primary Count or there is a problem. MARS guarantees that our counters are 99.6%. If the compare totals screen shows counts different by more than 0.4% for a counter, then please contact MARS to assist with fixing the problem.

Trolley Debug Data

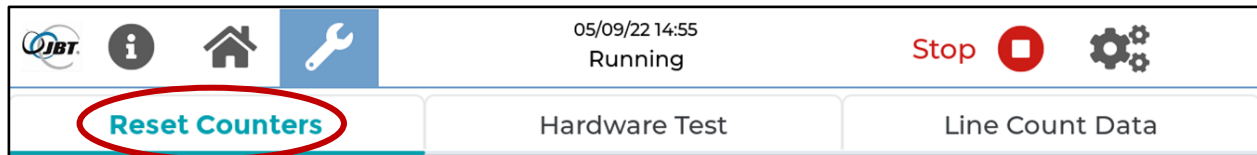
The Trolley Debug Data screen shows the 6 most recent Trolley block times for each counter, that exceeded the value configured for the Trolley Debug Min block time on the Line Count Setup screen.



The special Trolley Count trolley that blocks the shaft photoeye longer than the other trolleys only comes around once for each revolution of the chain. A complete revolution may take 10 to 15 minutes. It would be difficult and time-consuming to use the Hardware Test screen to accurately see how long this special trolley blocks the shaft photoeye.

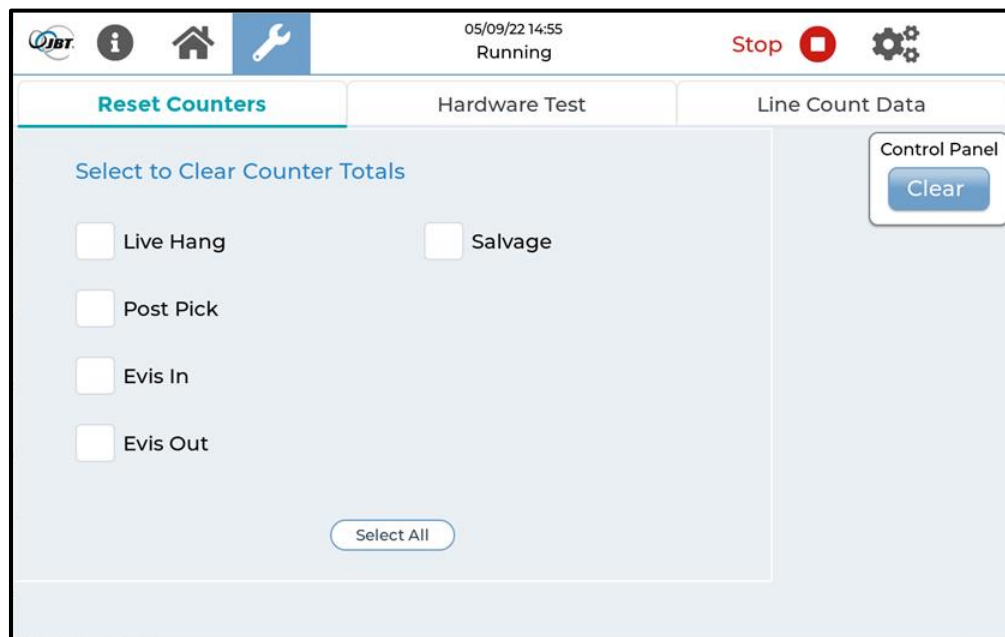
The Trolley Debug Data screen allows you to see the block times spread out over an hour or more. The values shown on this screen can then be used to accurately configure the Trolley Count minimum and maximum block times on the Line Count Setup screen.

Tools (Wrench) – Reset Counters



Navigation: Tools (Wrench) – Reset Counters (Upper Tab)

Security Level: Level 1 Operator Password or above.



The 'Clear' button appears after logging in. Touch individual counters to clear totals or 'Select All' to select all counters.

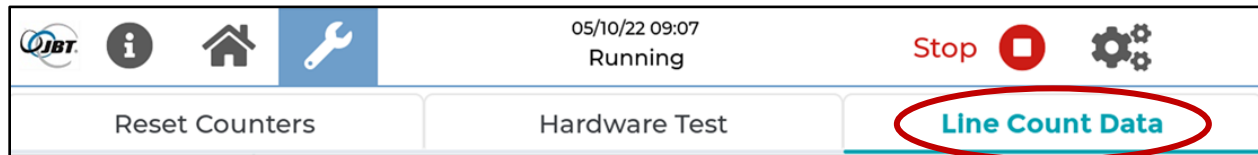
Checkbox touch areas include the label. Touching any part of the checkbox/label line is usually sufficient to toggle the checkbox.

After a Counter is selected, a red border appears around the 'Clear' button to alert the user that a change has been made to the screen, and the user must press the **Clear Button** to complete the process before leaving the screen.

OPERATIONS AND INSTRUCTIONS

LINE COUNT DATA

Tools (Wrench) – Line Count Data



Navigation: Tools (Wrench) – Line Count Data (Upper Tab)

Security Level: Level 1 Operator Password or above.

Note: This report is only accessible Trolley Count has been configured for the line.

The Line Count Data screen shows the 4 most recent Trolley Counts, Shackle Counts and Trolley Deltas. See Line Count Setup for more information.

The screenshot shows the 'Line Count Data' screen. It has the same top navigation bar as the previous image. Below the tabs, there is a 'Live Hang' section on the left and a table on the right. The table is titled 'Live Hang' and has four columns: 'Trolley Count', 'Shackle Count', 'Shackles Missing', and 'Trolley Delta'. The table contains four rows of data, all with the value '0'. Below the table, there is a red text note: 'Only counters with Trolley Count enabled are included on this report.'

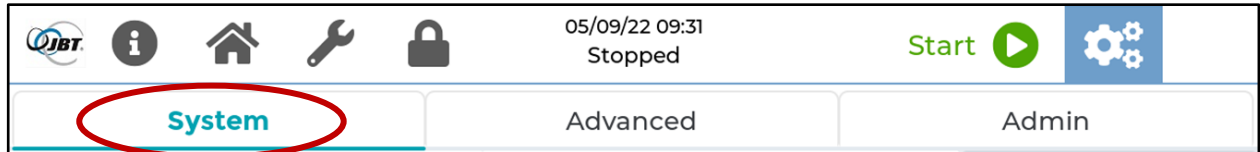
Trolley Count	Shackle Count	Shackles Missing	Trolley Delta
0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0

The Shackles Missing column is calculated by subtracting the Shackle Count from the Trolley Count. The Shackle Count is only used with Evisceration Lines and the Shackles Missing value indicates how many trolleys on the Evisceration Line are missing the shackle portion below that holds the bird. The shackle portion is typically made of plastic and can break off from the trolley.

OPERATIONS AND INSTRUCTIONS

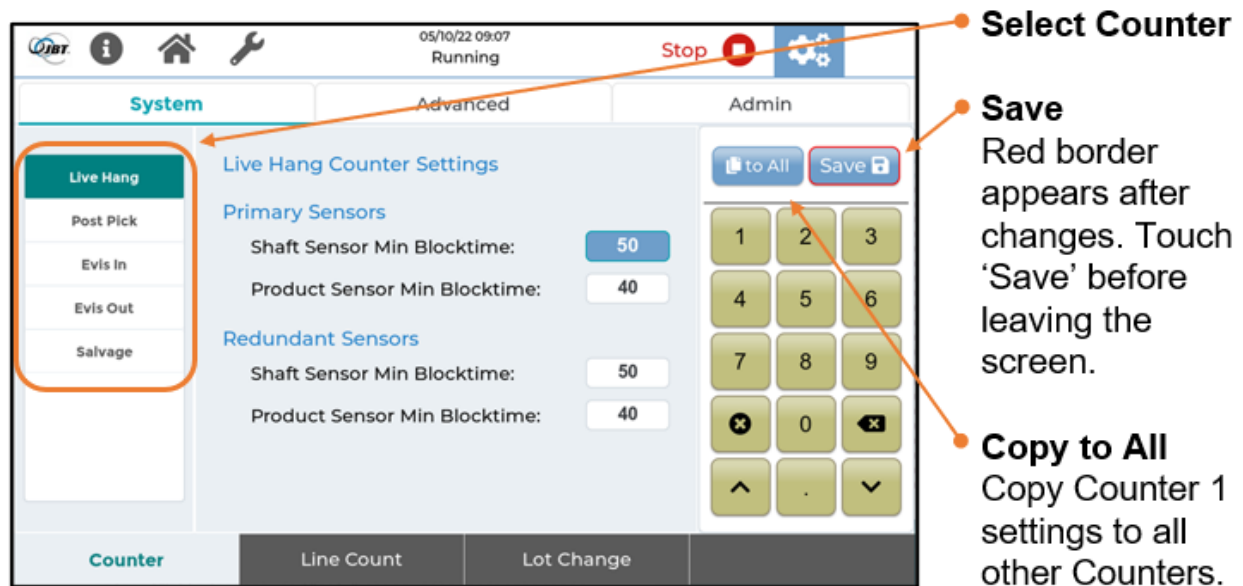
SYSTEM SETTINGS - COUNTER SETTINGS

Settings (Cogs) – System – Counter Settings



Navigation: Settings (Cogs) – System (Upper Tab) – Counter (Lower Tab)

Security Level: Level 2 System Password or above



Minimum Photoeye Block Times

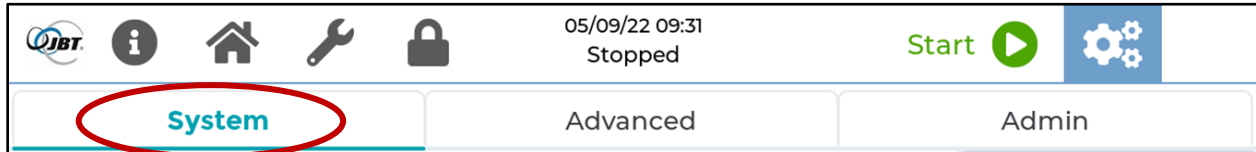
The minimum photoeye block time (milliseconds) required for a valid block.

- **Shaft sensors:** Minimum time the shaft photoeye must be blocked for the system to count a shaft (shackle).
- **Product sensors:** Minimum time the product sensor must be blocked, **while the shaft sensor is also blocked**, for the system to count a bird (product) on the shackle.

OPERATIONS AND INSTRUCTIONS

SYSTEM SETTINGS - LINE COUNT SETUP

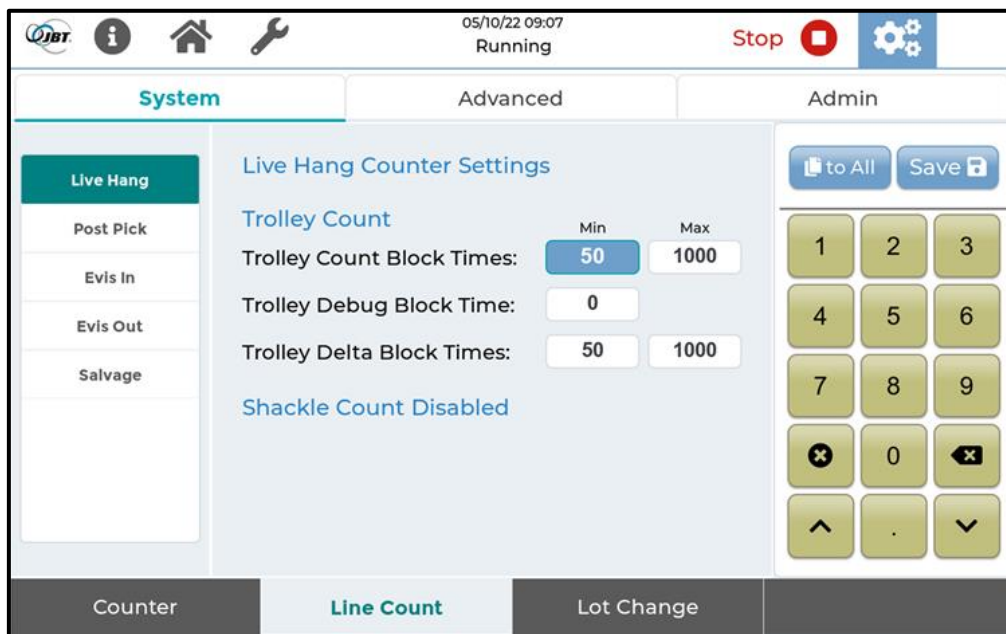
Settings (Cogs) – System – Line Count Setup



Navigation: Settings (Cogs) – System (Upper Tab) – Line Count (Lower Tab)

Security Level: Level 2 System Password or above.

Note: Only configured options are available for use.



Trolley Count: Minimum and maximum time that the shaft photoeye must be blocked by the special trolley that signals that a complete revolution of the chain has occurred. This special trolley has a piece of metal welded to it so that it blocks the shaft photoeye longer than the other trolleys. The Trolley Debug Min Block Time value below along with the Trolley Debug Data (Advanced Setup Menu) can be used to determine how long the special trolley blocks the shaft photoeye.

Trolley Delta: Used to Count the number of trolleys from one point on a line to another by reading a second trolley which has been designed to have a longer block time.

OPERATIONS AND INSTRUCTIONS

SYSTEM SETTINGS - LINE COUNT SETUP

Shackle Sensor: Minimum and maximum time that the special shackle photoeye on the Evisceration Line must be blocked to count a valid trolley with a shackle intact below it. The Hardware Test screen can be used while the Evisceration Line is running to observe typical shackle block times.

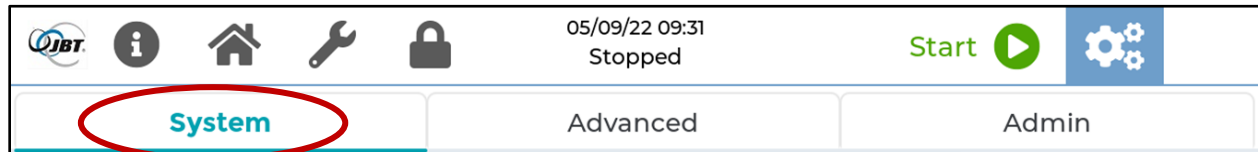
Shackles to Ignore: This value is usually set to 0 to specify that every valid block of the shackle photoeye discussed above will count toward the Shackle Count. Some plants may have chains that cause the shackle photoeye to block a second time when every shackle goes by. In this case this value would be set to 1 specifying that only every other valid block of the shackle photoeye should count toward the Shackle Count.

Trolley Debug Min Blocktime: This value should be set to slightly less than how long the special Trolley Count trolley will block the shaft photoeye. This value should also be set higher than how long a typical trolley would block. The special trolley for the Trolley Count that indicates a whole revolution of the chain has occurred will block longer than a typical trolley because of the extra metal that has been welded to it. The actual block time will be stored in memory for the last 6 trolleys that have gone by where the block time exceeded this setting. The 6 most recent trolley block times that exceeded this setting can then be accessed on the Trolley Debug Data screen from the Advanced Setup menu. The 6 values shown can then be used to set the Trolley Debug Min Block time correctly.

OPERATIONS AND INSTRUCTIONS

SYSTEM SETTINGS - LOT CHANGE SETUP

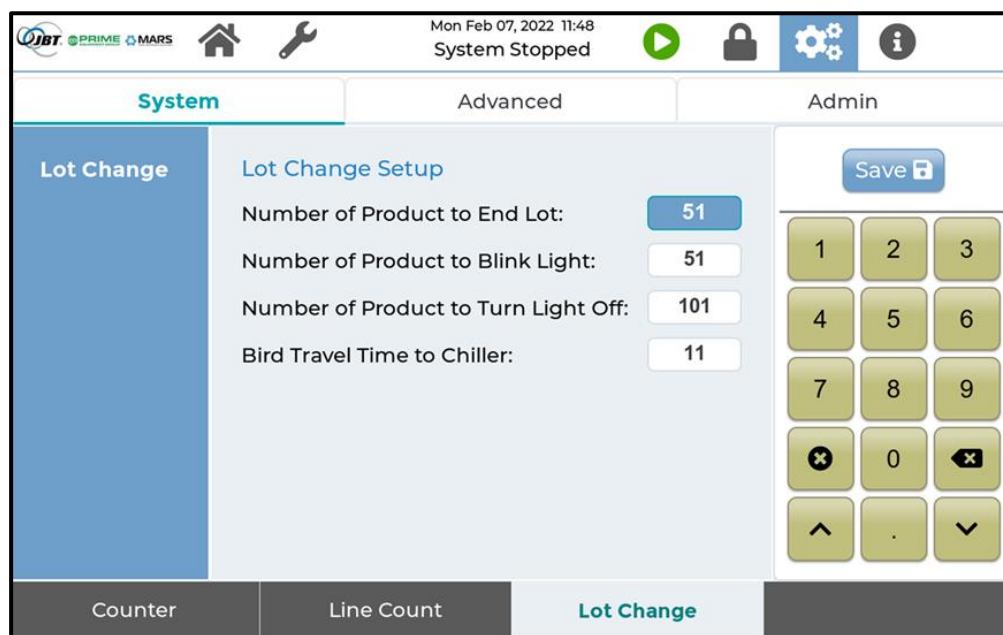
Settings (Cogs) – System – Lot Change Setup



Navigation: Settings (Cogs) – System (Upper Tab) – Lot Change (Lower Tab)

Security Level: Level 2 System Password or above

Note: Lot Change Setup only available if Lot Change is configured for your system.



The **Lot Change** settings are used to configure what happens when a lot change is occurring. Lot Change is used to get a total count of birds from a particular grower so the total count can be compared against the grower's count.

- Press the Lot Change Button for 3 Seconds to mark the End of a Lot.
- Press the Lot Change Button for 30 Seconds to mark the End of Shift.

When all product have been unloaded from a particular grower, press the Lot Change button for 3 seconds. The JBT First Processing Counter System will close out the current lot and start a new lot. The total bird count for the lot being closed will be saved and uploaded to the JBT Efficiency Management System running on a Windows computer. The Lot Change data will then be available in the Lot Change report under the Counters tab.

OPERATIONS AND INSTRUCTIONS

SYSTEM SETTINGS - LOT CHANGE SETUP

Number or Product to End Lot: How many product must be counted after the Lot Change button is pressed for 3 seconds to actually close out the lot, save the total count for the lot and start a new lot count. Set this value to the number of live birds that are typically still on the belt when the Lot Change button is pressed.

Number of Product Light Blink: How many product must be counted after the Lot Change button is pressed to start blinking a light notifying personnel in the plant that a lot change is in progress. Set this value to the number of shackles from the picking line live hang counters to the rehang transfer point, which should be a few shackles past the picking line post pick counters.

Number of Product Light Off: How many product must be counted before the light will stop blinking indicating that the lot change is complete. Set this value to the number of shackles from the picking line live hang counters to the USDA station on the evisceration line.

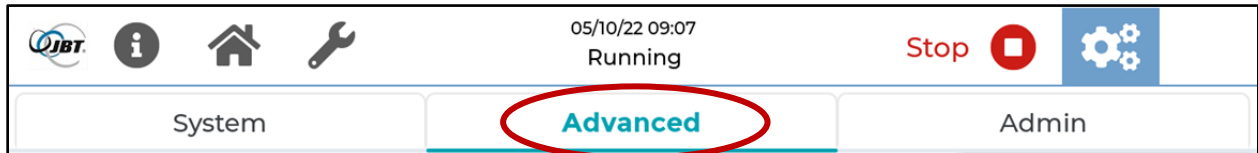
Save Button

Red border appears after changes. Touch 'Save' before leaving the screen.

Copy to All Button

Copy Counter 1 settings to all other Counters.

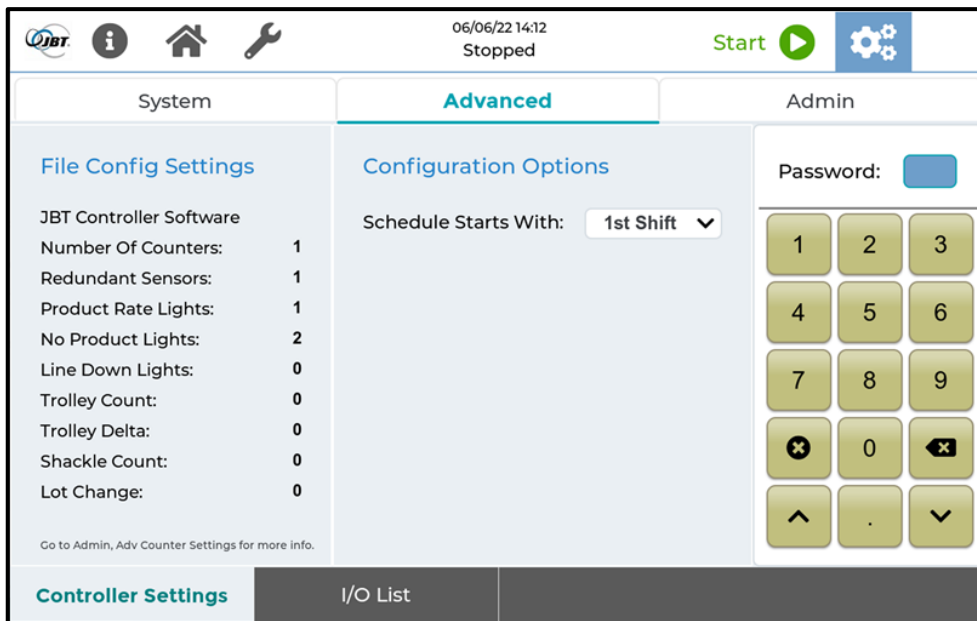
Advanced Settings – Controller Settings



Navigation: Settings (Cogs) – Advanced (Upper Tab) – Controller Setup (Lower Tab)

Security Level: Level 3 Advanced Password or above

These values are usually set during the initial installation.



File Config Settings: Configuration settings that are specified in your plant's configuration file. This section is for reference only and provides information about how your system is configured.

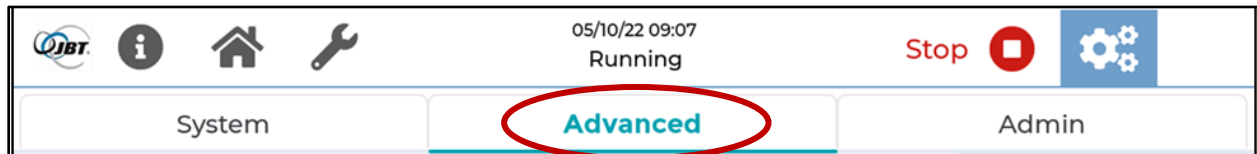
Configuration Options

Schedule Starts With: Specify whether the plant's day begins with first or 3rd shift.

OPERATIONS AND INSTRUCTIONS

ADVANCED SETTINGS - I/O LIST

Advanced Settings – I/O List



Navigation: Settings (Cogs) – Admin (Upper Tab) – I/O List (Lower Tab)

Security Level: Level 3 Advanced Password or above

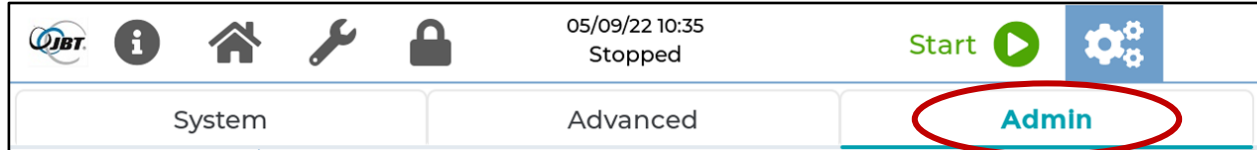
The I/O (Inputs/Outputs) screen provides a convenient list of the inputs and outputs configured for your plant's installation. Testing these inputs/outputs is performed on Hardware Test.

The screenshot shows the 'Advanced' tab selected in the JBT interface. Below the tabs, there is a table with four columns: 'Inputs', 'Inputs Cont', 'Outputs', and 'Outputs Cont'. The table lists various inputs and outputs with their corresponding line numbers. At the bottom of the table, there are two buttons: 'Controller Settings' and 'I/O List'.

Inputs		Inputs Cont		Outputs		Outputs Cont	
Line 1 Shaft Primary	1	L3 Prod Primary	9	L1 Prod Rate Green	21		
L1 Prod Primary	2	L4 Shaft Primary	10	L1 Prod Rate Red	22		
L1 Shaft Redundant	3	L4 Prod Primary	11	L1 No Prod Primary	23		
L1 Prod Redundant	4	L5 Shaft Primary	12	L1 No Prod Redundant	24		
L1 Lot Change Btn	5	L5 Prod Primary	13	L1 Lot Change Light	25		
L2 Shaft Primary	6			L2 No Prod Primary	26		
L2 Prod Primary	7			L3 No Prod Primary	27		
L3 Shaft Primary	8			L4 No Prod Primary	28		
Controller Settings		I/O List					

Your I/O List will be different than the sample shown here.

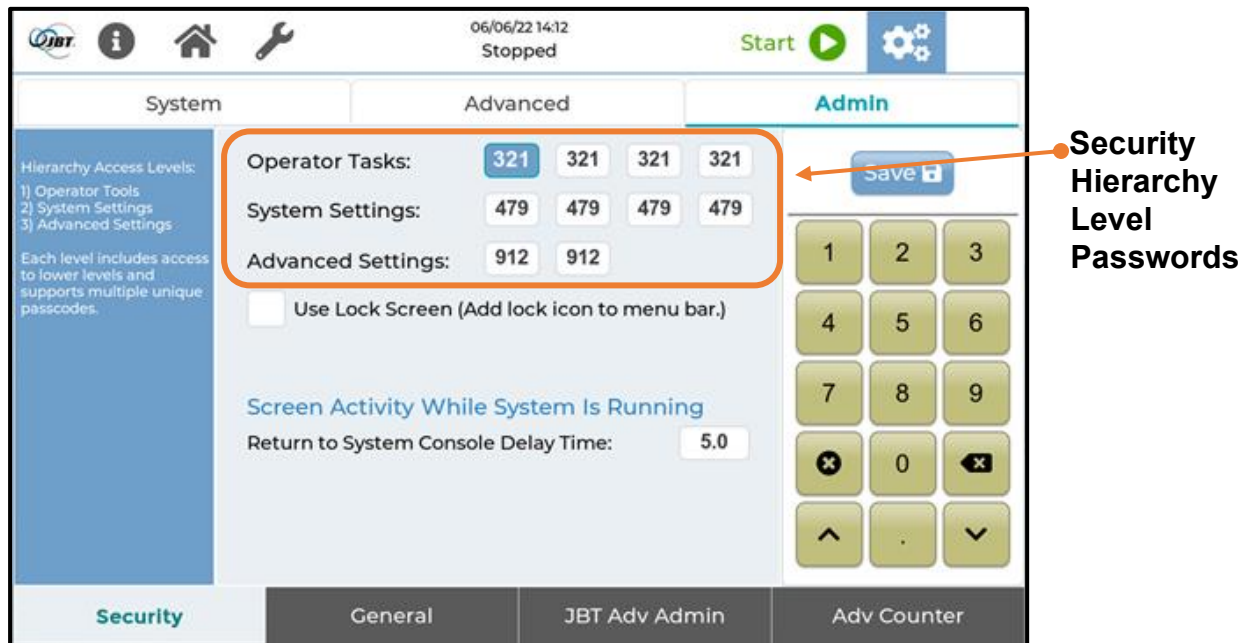
Admin Screens – Security Setup



Navigation: Settings (Cogs) – Admin (Upper Tab) – Security (Lower Tab)

Security Level: Level 3 Advanced Password

The Gen 2 Security has changed from Gen 1's individual passwords managed externally, to a **hierarchy style managed by the plant**.



Security Hierarchy

Passwords are evaluated by security level, and higher level passwords work on lower level tasks. For example, an Operator password is only valid on Operator tasks, were as System level password works on both Operator and System levels, and Advanced passwords work on Operator, System and Advanced tasks.

Level 1 Operator passwords are required for Operator type activity such as enabling reject buttons

Level 2 System Setting passwords are required for system configuration changes. These passwords should only be given to those who have the knowledge and authority to change system settings. Level 2 password also work for Level 1 Tasks.

Level 3 Advanced Setting passwords are required for advanced system settings that should only be done by trained individuals. Level 3 passwords also work for Level 1 & Level 2 Tasks.

Return to System Console Delay Time

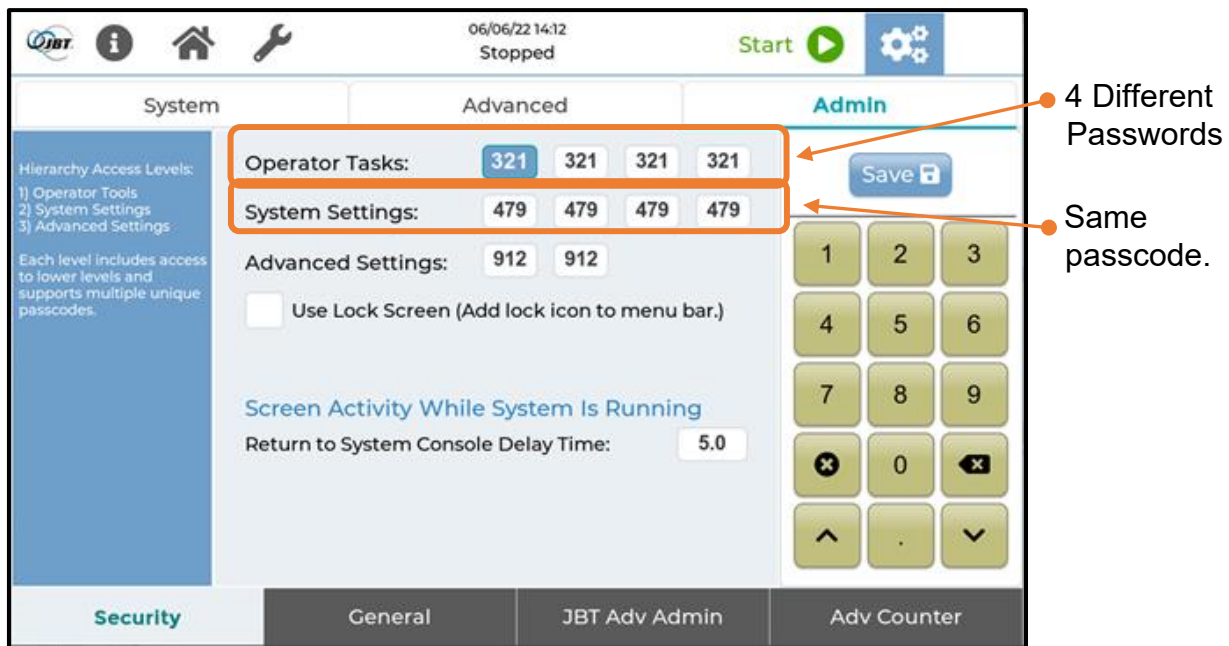
If the system is running, the home screen/system console is not the active screen, and there has been no user activity, the system will return to the home screen after the configured amount of time has passed.

OPERATIONS AND INSTRUCTIONS

ADMIN SCREENS - PLANT MANAGED PASSWORDS

Plant Managed Passwords

Gen 2 provides a designated login to manage passwords at the plant level. The Security Admin password only works on the Security Screen and can't be used to perform system tasks. Operator and System levels support up to 4 different passwords, and Advanced Level support 2 passwords.



Invalid Passwords

Passwords must be three digits and can not start with 0. An invalid password entry will be changed back to the original value and a red border added to the textarea block to alert the user that the entered password was rejected.

When to Assign 4 Different Passwords

The Operator Tasks example above shows four different passwords which can be given to the various operators as needed.

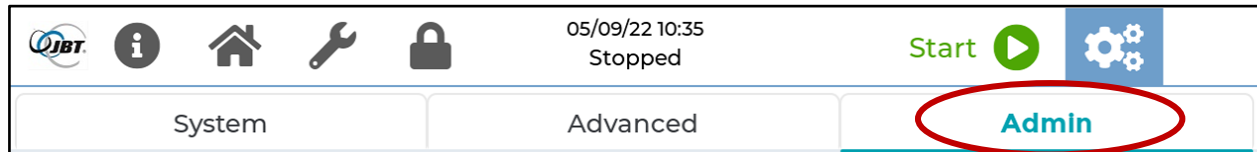
When to Assign Same Passwords

For plants transitioning from Gen1 to Gen2, it's possible that only one or two people need a system level password, or everyone is used to using a certain password. **Using the same password allows the plant to continue with their Gen1 password and assign other passwords at their convenience.**

OPERATIONS AND INSTRUCTIONS

ADMIN SCREENS - GENERAL

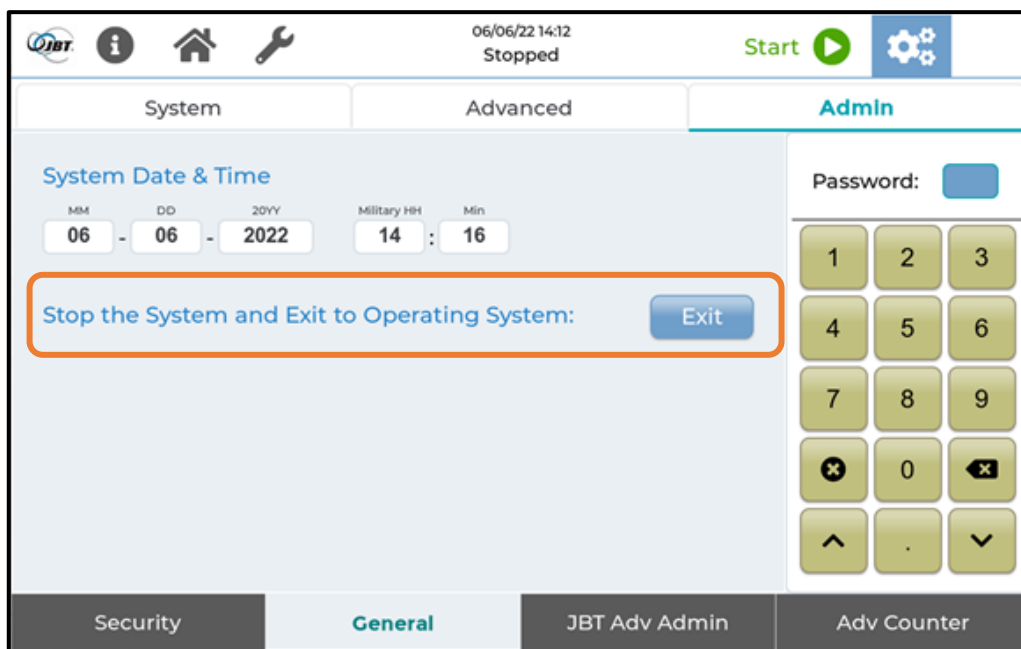
Admin Screens - General



Navigation: Settings (Cogs) – Admin (Upper Tab) – General (Lower Tab)

Security Level: Level 3 Advanced Password or above

The General Admin screen contains settings and tools that are unrelated to your system's configuration or operation.



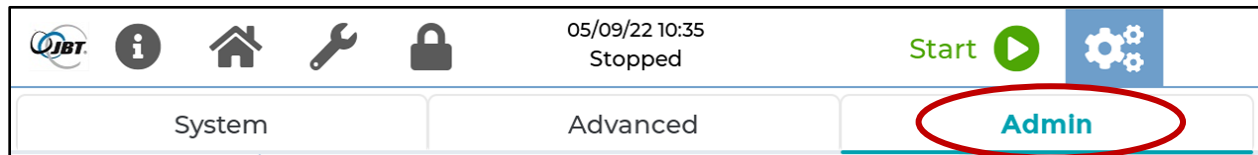
System Date & Time: The JBT controller has a battery-backed up clock for maintaining the correct Date and Time when the controller is not connected to the internet. Enter the correct Date and Time after logging in with your password. Then press the Save button.

Exit to Operating System: Stop the system and exit the JBT software to the Operating System.

OPERATIONS AND INSTRUCTIONS

ADMIN SCREENS - JBT ADVANCED ADMIN

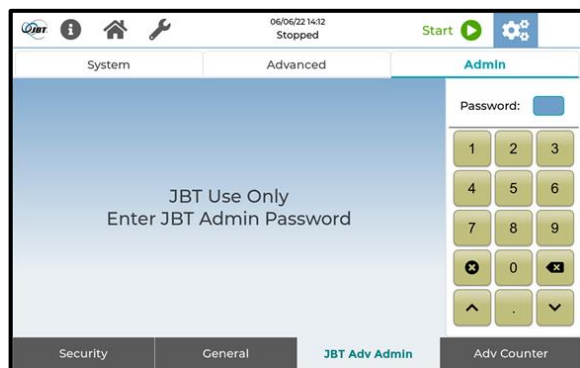
Admin Screens – JBT Advanced Admin



Navigation: Settings (Cogs) – Admin (Upper Tab) – JBT Adv Admin (Lower Tab)

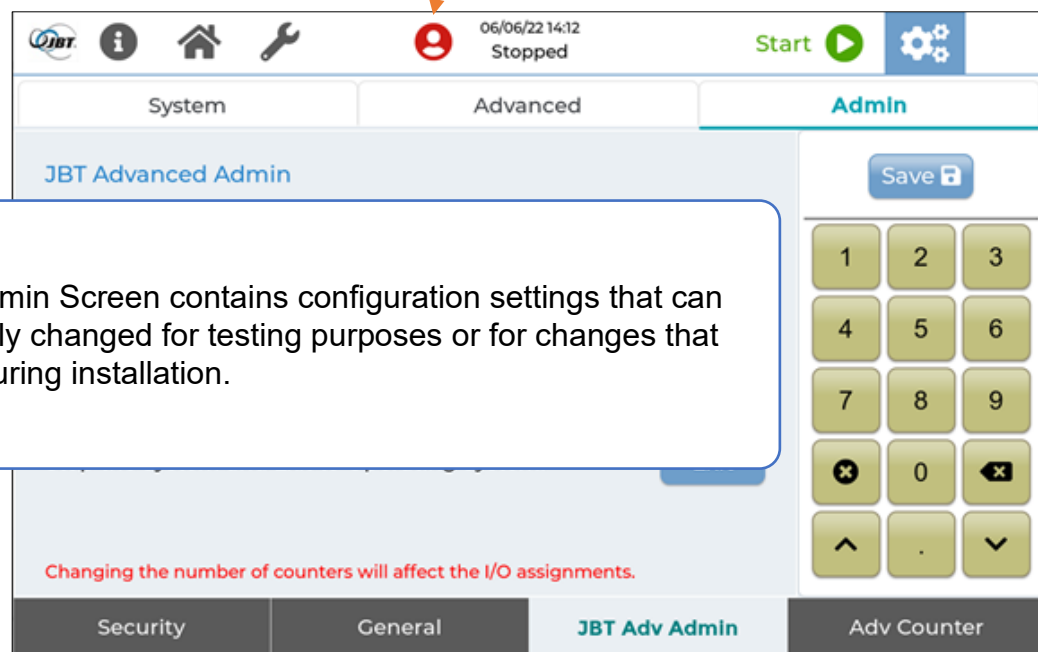
Security Level: JBT Super User (JBT Use Only)

The JBT Admin screen contains settings and tools that are used exclusively by a JBT Technician during installation.



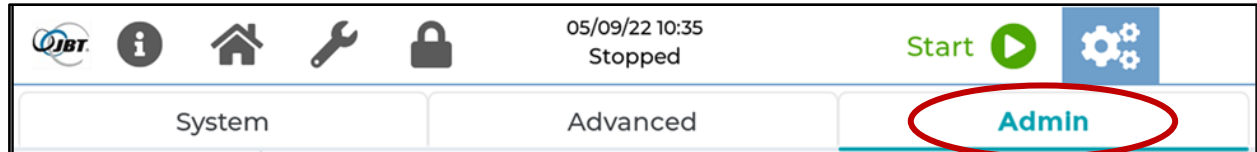
JBT Admin is logged in and passwords are disabled.

Touch icon to logout as JBT Admin.



JBT Admin Screen contains configuration settings that can be easily changed for testing purposes or for changes that arise during installation.

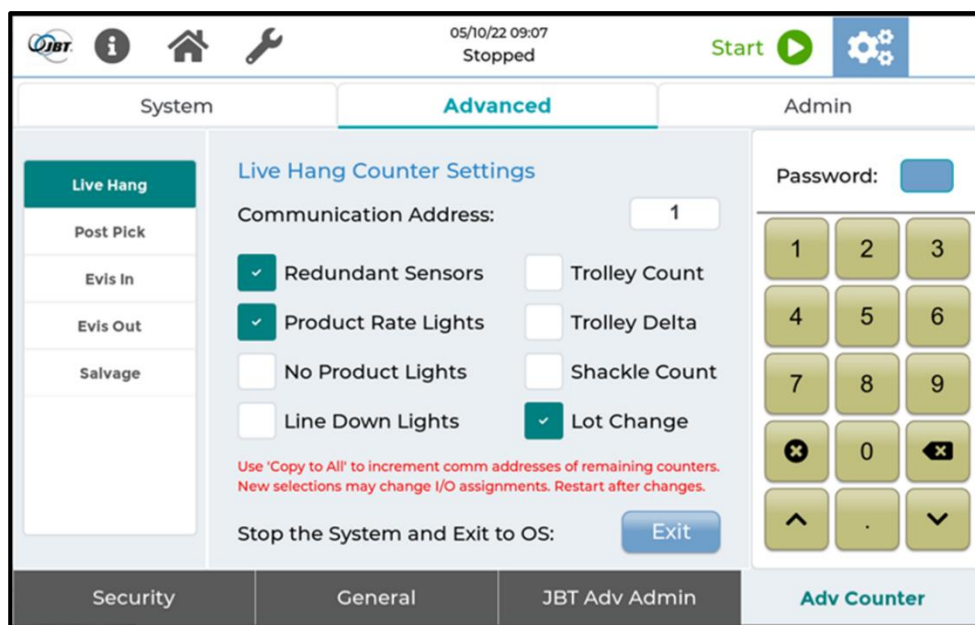
Admin Screens – Advanced Counter Setup



Navigation: Settings (Cogs) – Admin (Upper Tab) – Adv Counter (Lower Tab)

Security Level: JBT Technician

Installation: The Advanced Counter screen is used by a MARSJBT Technician during the installation process to configure the counter options for each line. Most modifications will affect the I/O assignments and the hardware used.



Post-Installation: The Advanced Counter screen is visible as a setup reference. Select the counter names on the left to view settings per counter.

Advanced Counter Settings

Redundant Sensors: If selected, an additional set of photoeye sensors will be used on the selected line, and two separate shackle counts and product counts will be available for comparison.

Product Rate Lights: Green and Red Lights are installed where the operators can see them while working

- **Green Light:** The green light will turn on when the product count for the last minute met the desired target rate.
- **Red Light:** The red light will turn on when the product count for the last minute was less than the desired target rate.

The lights are off during break times and after production has finished for the day.

No Product Light: The No Product Light is used to verify that the counter hardware and software is detecting empty shackles or cones properly. The no product light will come on every time a shackle goes by without product.

Line Down Lights: Green and Red Lights are installed to quickly alert production supervisors and maintenance personnel when a line has stopped.

- **Green Light:** The line is running.
- **Red Light:** The line has stopped.

The Green and Red lights can be installed anywhere; however, they are often installed in the production supervisor's office or in the maintenance shop.

Check the appropriate setting if the system hardware has been configured to support these features:

- **Trolley Count**
- **Trolley Delta**
- **Shackle Count**
- **Lot Change**

SANITATION

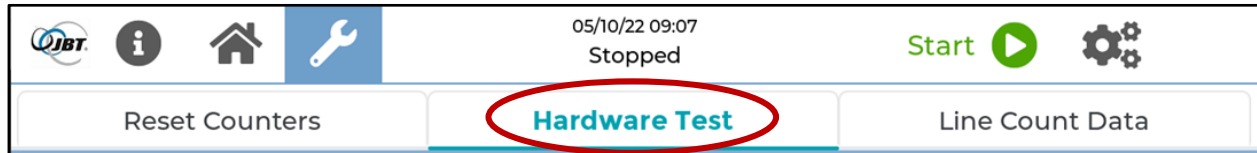
SANITATION PROCEDURES

There are no sanitation procedures for this manual.

MAINTENANCE AND TROUBLESHOOTING

HARDWARE TEST

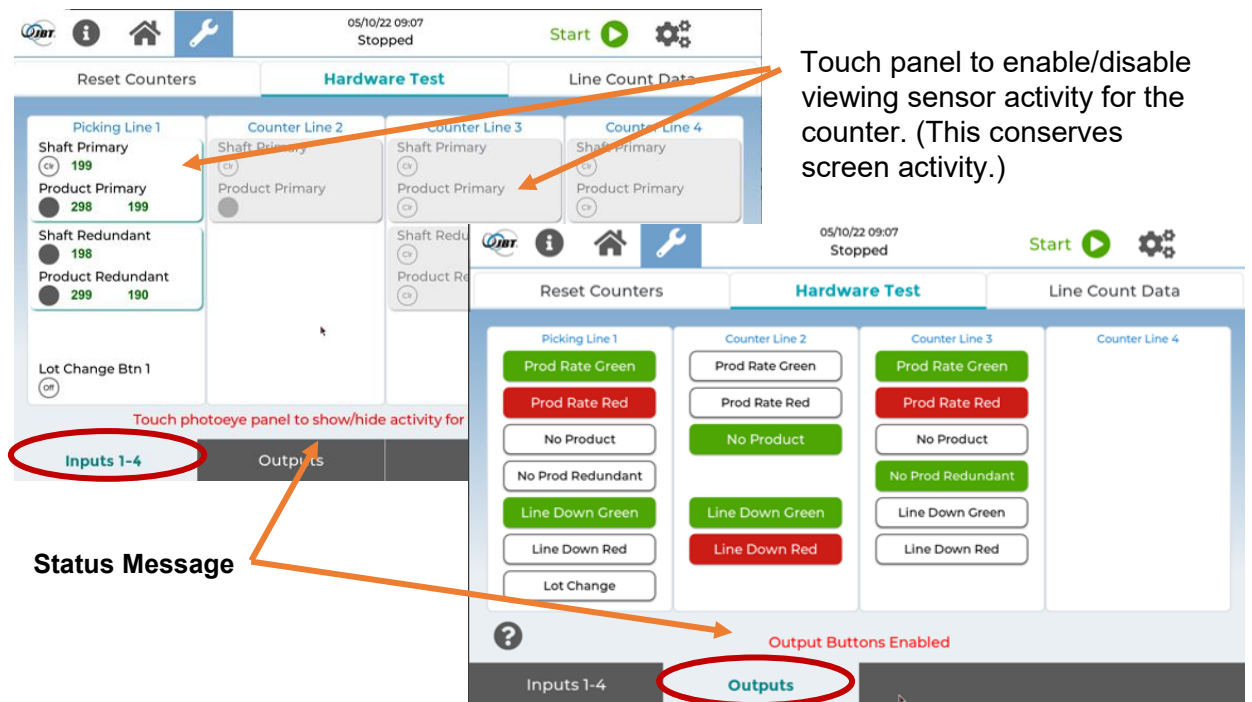
Tools (Wrench) – Hardware Test



Navigation: Tools (Wrench) – Hardware Test (Upper Tab)

Security Level: Level 2 System passcode or above

Your Hardware Test screen will reflect the inputs and outputs for your system.



Verify Inputs

The input buttons are shown as small circles which reflect the current state of the input.

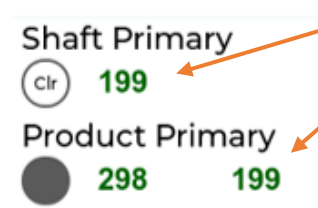
Pushbutton States:  

Photoeye Sensors:  

Product Photoeye Block Times

The amount of time the photoeye was blocked by product.

The amount of time the product photoeye was blocked while the corresponding shaft or cone photoeye was also blocked.

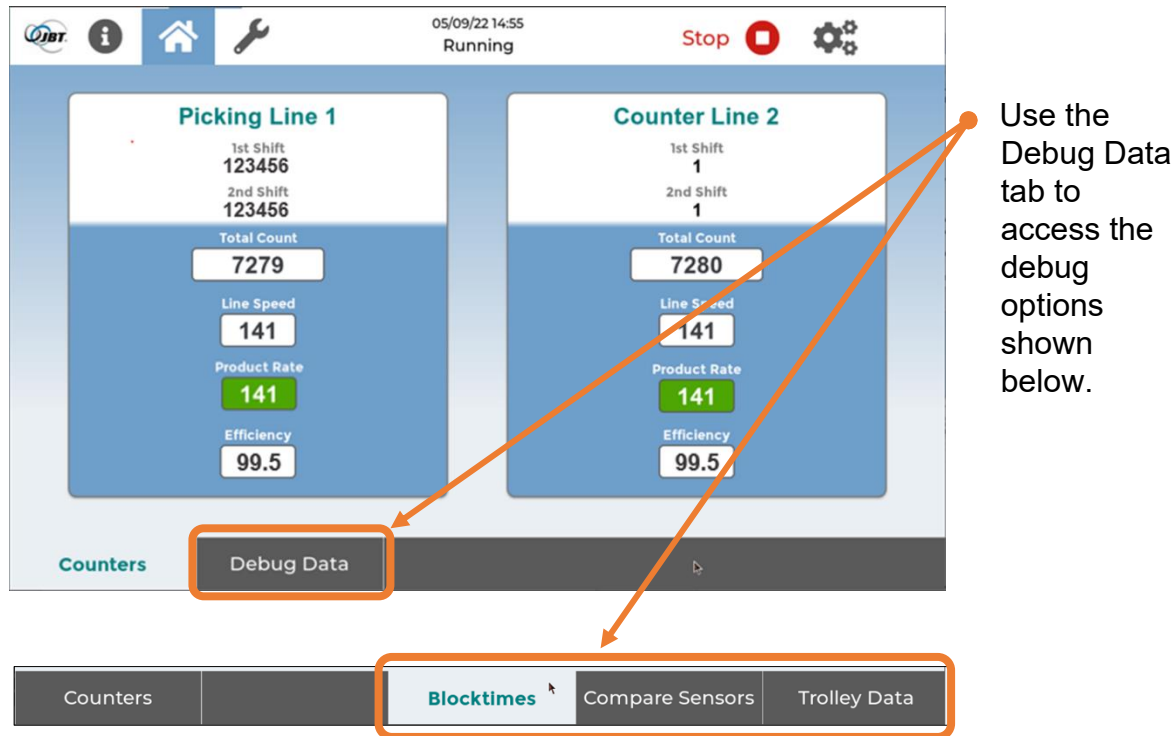


MAINTENANCE AND TROUBLESHOOTING

DEBUG DATA

Debug Data

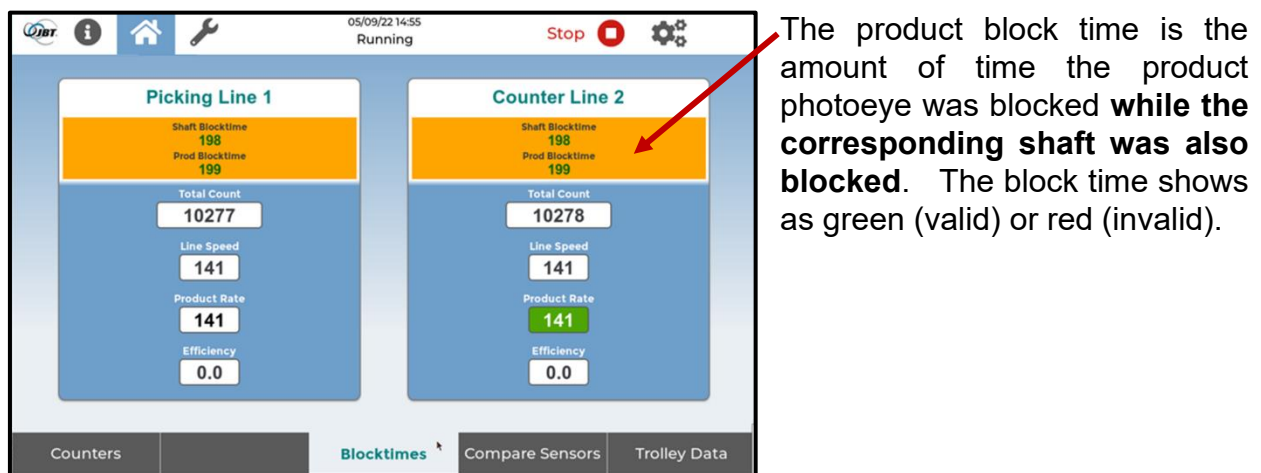
The **Debug Data** tab provides a login to access debug data screens.



Touching the Counters tab or the Home icon cancels the Debug Screens.

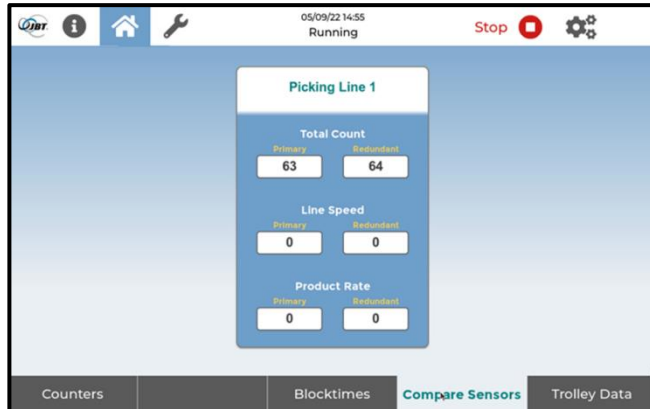
Show Block Times

The block time durations are shown in the yellow section below.



Compare Sensors

Compare the counts of the primary and redundant sensors. This feature only applies to counters with redundant sensors.

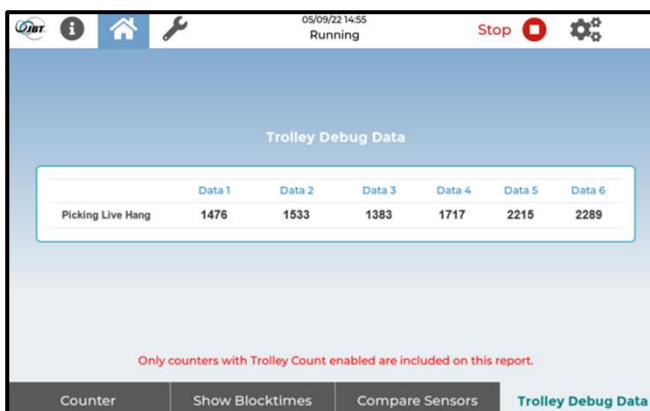


For each counter with redundant sensors, the **Primary Count** is displayed with the **Compare Count** below. The Primary Count is the current product (bird) count for the primary sensors and the Compare Count is the current product count for the redundant sensors.

Typically, the Redundant Count should be within 0.4% of the Primary Count or there is a problem. MARS guarantees that our counters are 99.6%. If the compare totals screen shows counts different by more than 0.4% for a counter, then please contact MARS to assist with fixing the problem.

Trolley Debug Data

The Trolley Debug Data screen shows the 6 most recent Trolley block times for each counter, that exceeded the value configured for the Trolley Debug Min block time on the Line Count Setup screen.



The special Trolley Count trolley that blocks the shaft photoeye longer than the other trolleys only comes around once for each revolution of the chain. A complete revolution may take 10 to 15 minutes. It would be difficult and time-consuming to use the Hardware Test screen to accurately see how long this special trolley blocks the shaft photoeye.

The Trolley Debug Data screen allows you to see the block times spread out over an hour or more. The values shown on this screen can then be used to accurately configure the Trolley Count minimum and maximum block times on the Line Count Setup screen.

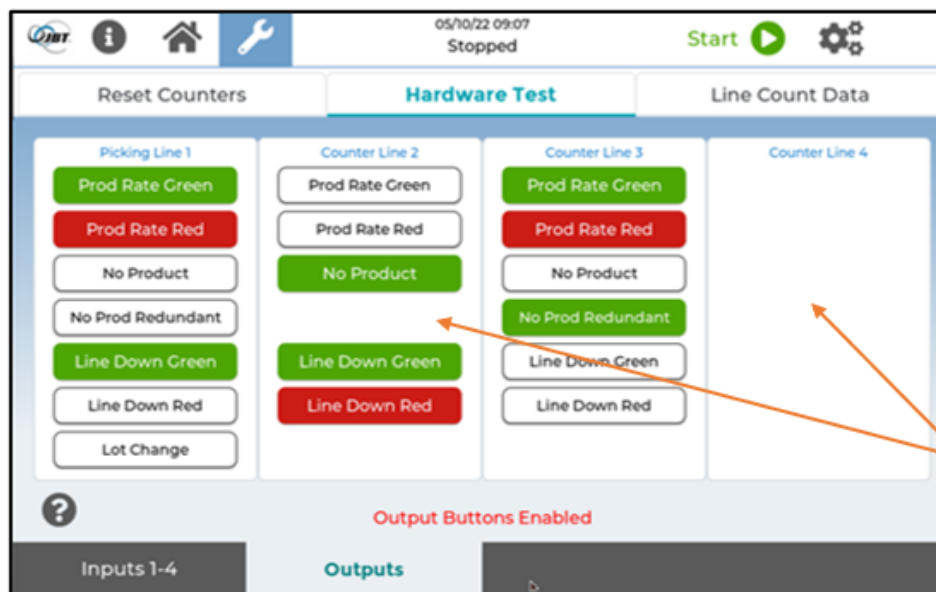
MAINTENANCE AND TROUBLESHOOTING

TEST OUTPUTS

Test Outputs

Use the output buttons to test the signal used to control hardware such as gates and external lights. The output buttons also reflect the state of the output while the system is running. Touch the appropriate button to verify the correct response.

See Settings (Cogs) – Advanced (Upper Tab)- I/O List (Lower Tab) for I/O assignments.



The 1st Processing Counter system has many output options that vary by plant configuration and counter.

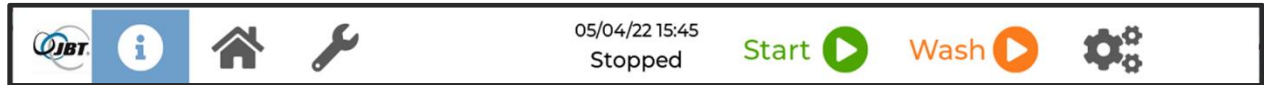
Most configurations will only have a few, if any, output options per counter.

Unused output options are blank on the Outputs screen.

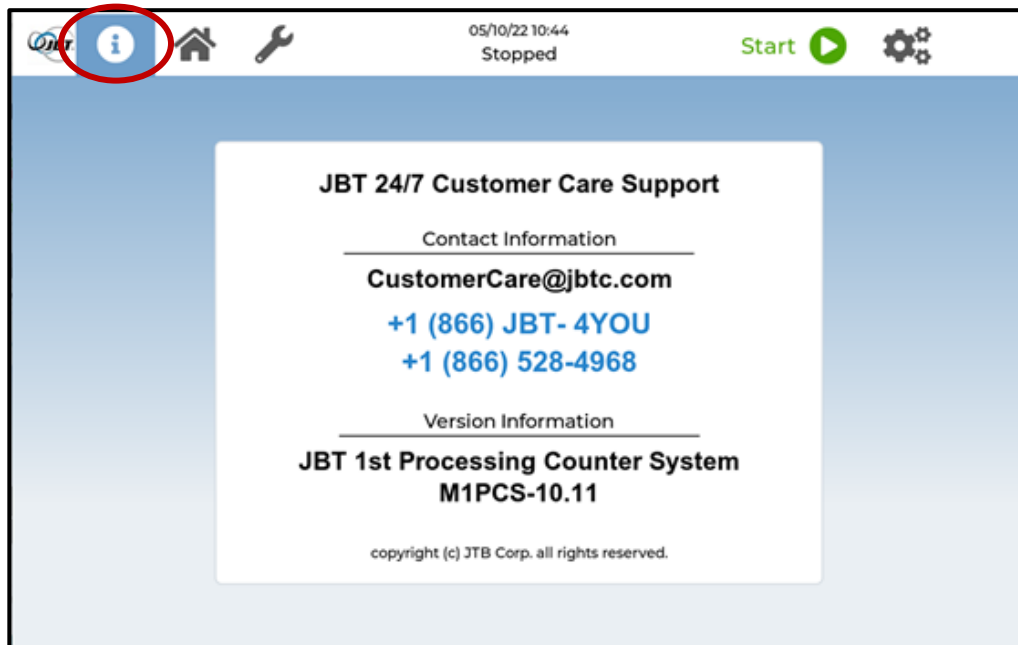
Touch the Output Buttons on the screen to manually test the lights installed for our system.

Output	System Activity
Product Rate Green Light	Turned on when product rate for the last minute is at the desired rate.
Product Rate Red Light	Turned on when product rate for the last minute is slower than desired.
No Product Lights	Turned on and off when a shackle without product passes by the primary/redundant photoeye sensors.
Line Down Green light	Turned on when the line is running.
Line Down Red light	Turned on when the line is stopped.
Lot Change light	Turned on when a lot change is in process. Turned off when the lot change is complete.

Info and About



Company, Contact, and Software Version Information



Contact Info

Phone: 1 - 866 – JBT – 4YOU
1 - 866 – 528 - 4968
Email: CustomerCare@jbtc.com

Contact a JBT Service Technician

If you have a question about your **JBT System** or require service after trying to troubleshoot the issue, please contact us using the contact information provided on your system's Info screen.

Troubleshooting

The **Hardware Test** screen is a good place to start when experiencing problems with the correct operation of your **JBT System**.

If all the external hardware appears to be functioning properly on the **Hardware Test** screen, then diagnosing the problem may require assistance from a JBT service technician.

ASSEMBLY DRAWINGS

The following pages include the Bill of Materials and Assembly drawings for

CONTROLS AND ELECTRICAL SCHEMATICS

PART NO.	DESCRIPTION

SPARE PARTS LIST

To order spares call 866.528.4968

Email: parts.prime@JBTC.com

Refer to serial number:

Refer to model number:

Although many parts may seem similar between 2 machines, due to wear and adjustments they become machine specific. You will cause damage to your machine if you swap parts with other machines.

PART NO.	DESCRIPTION	QTY.
11825	PWRSPLY,100-550VAC,24VDC*	1
71912	BOARD, CNTRL, NEXT GEN	1
80742	DISPLAY, 10", PNL MT, ALUM BEZEL, IP65	1
80787	CBL, HDMI, 3FT	2
MRS1241	LIGHT,30MM,10-30V,G-R-Y,2M CABLE	1
MRS1281	3 Color Light Assembly	1
MRS6001	Hardwire Send Sensor w/30' Cable	2
MRS6011	Hardwire Receive Sensor w/30' Cable	2

TECHNICAL PARTS INFORMATION

PART NO.	DESCRIPTION

